

Workshop 2: Matrix Algebra

1. *Review.* Determine the matrix for the transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that projects vectors vertically onto the line $y = 2x$. Also show that this transformation but with the line $y = 2x + 3$ is not linear.

$$\begin{bmatrix} x \\ y \end{bmatrix} \mapsto \begin{bmatrix} x \\ 2x \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \mapsto \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \mapsto \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\vec{x} \mapsto \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} \vec{x}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} \mapsto \begin{bmatrix} x \\ 2x+3 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} \mapsto \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

This transformation doesn't map $\vec{0}$ to $\vec{0}$

2. Determine if the transformation $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} x_1 \\ x_2 \\ x_1 + x_2 \end{bmatrix}$ is one-to-one, onto, both or neither. Justify your answer.

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \mapsto \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \mapsto \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \sim \begin{bmatrix} \boxed{1} & 0 \\ 0 & \boxed{1} \\ 0 & 0 \end{bmatrix}$$

2 pivots $\Rightarrow$ 1-to-1
2 columns

2 pivots $\Rightarrow$ not onto
3 rows

3. Determine if the transformation $\mathbf{x} \mapsto \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \mathbf{x}$ is one-to-one, onto, both or neither. Justify your answer. Also draw the effect of this transformation on the unit square.

$$\begin{bmatrix} \boxed{1} & 0 \\ 0 & 0 \end{bmatrix}$$

1 pivot $\Rightarrow$ not 1-to-1
2 columns

1 pivot $\Rightarrow$ not onto
2 rows

4. Determine a general form solution for the equation $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 0 & 3 \\ 1 & 6 \end{bmatrix} \mathbf{x}$.

$$A\vec{x} = B\vec{x} \Leftrightarrow (A-B)\vec{x} = \vec{0}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - \begin{bmatrix} 0 & 3 \\ 1 & 6 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 2 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$$

$x_1 = x_2$
 x_2 is free

5. Let $A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \\ 3 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ -1 & 4 \\ 1 & 1 \end{bmatrix}$. Calculate AB . If it's not possible, then explain why.

$3 \times \boxed{2} \quad \boxed{3} \times 2$
don't match

Not possible

6. Let A and B be as in the previous problem. Calculate $((A^T B)^2)^T$. If it's not possible, then explain why.

$$A^T B = \begin{bmatrix} 1 & 0 & 3 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ -1 & 4 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} (2+3) & (0+3) \\ (4+1+1) & (-4+1) \end{bmatrix} = \begin{bmatrix} 5 & 3 \\ 6 & -3 \end{bmatrix} \quad (A^T B)^T = \begin{bmatrix} 4 & 3 & 12 \\ 6 & 27 \end{bmatrix}$$

$$(A^T B)^2 = \begin{bmatrix} 5 & 3 \\ 6 & -3 \end{bmatrix} \begin{bmatrix} 5 & 3 \\ 6 & -3 \end{bmatrix} = \begin{bmatrix} (25+18) & (15-9) \\ (30-18) & (18+9) \end{bmatrix} = \begin{bmatrix} 43 & 6 \\ 12 & 27 \end{bmatrix}$$

7. Determine two matrices $A, B \in \mathbb{R}^{2 \times 2}$ such that $AB \neq BA$. Try doing this by thinking of two transformations S and T such that $S \circ T \neq T \circ S$ (where $S \circ T$ is S composed with T).

S = rotation by 90°

T = reflection across x -axis

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

8. Suppose that $A \in \mathbb{R}^{m \times n}$ and $(A^T B)^T C \in \mathbb{R}^{a \times b}$ (where all products are well-defined). Determine the shapes of B and C .

$$A^T \in \mathbb{R}^{n \times m}$$

$$B \in \mathbb{R}^{q_1 \times q_2} \quad C \in \mathbb{R}^{q_3 \times q_4}$$

$$A^T B \in \mathbb{R}^{n \times q_2} \Rightarrow q_1 = n$$

$$(A^T B)^T \in \mathbb{R}^{q_2 \times n}$$

$$(A^T B)^T C \in \mathbb{R}^{a \times b} \Rightarrow q_2 = a, q_3 = n, q_4 = b$$

$$B \in \mathbb{R}^{n \times a}$$

$$C \in \mathbb{R}^{n \times b}$$

9. Determine two matrices $A, B \in \mathbb{R}^{2 \times 2}$ such that $AB = 0$ but $A \neq 0$ and $B \neq 0$.

S = projection onto x -axis

T = projection onto y -axis

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$